

Abstract

In the Netherlands, growing concerns about environmental and animal welfare impact have driven livestock farming toward more sustainable farming system. However, the effects of the farming transitions on zoonotic diseases remain unclear. In this study, we used poultry farms and highly pathogenic avian influenza (HPAI) as a case study. Sustainable transitions were defined to reflect current trends in the Netherlands, including an increased of outdoor farming systems, a reduction in farm size, and a decrease in the number of farms. We assessed the effects of these changes on the risk of HPAI introduction from the environment into poultry farms, outbreak size, the reproduction number, and human exposure using a spatially explicit between-farm transmission model.

Outdoor farming increases the introduction risk of HPAI and thus more outdoor farms lead to a higher expected risk of HPAI introduction into farms. The risk was higher on farms located closer to water sources and grassland, as this increased the likelihood of contact with waterfowls. From the between-farm transmission model, a higher percentage of outdoor farming resulted in larger expected outbreak sizes, higher reproduction numbers, and higher expected human exposure to HPAI. Outdoor layer farms had a greater effect on outbreaks than outdoor broiler farms due to their higher susceptibility. The spatial pattern of outdoor farms highly affected outbreak size as the clustered outdoor farms sharply increased the expected outbreak size. Reducing farm size decreased the expected outbreak size, but was not sufficient to neutralize the effect of outdoor farms. Decreasing the number of farms also reduced the expected outbreak size. Changes in farming practices in the transition to sustainability could impact zoonotic outbreaks. Therefore, the transition should deliberately proceed with consideration of both its potential benefits and drawbacks.